Corvin — Project Budget

RackBotGen3 · Delta Cup 2026 · Team Boys Go to Thailand, Brawijaya University · spec v0.1.30 · 26 Oct 2026

1.0 Spending and forecast (25 Oct 2026, at Rp 18,000 = 1 USD)

What the project actually costs, from the Tokopedia cart total and the quotes received. §1.1–1.4 below are the itemised planning estimates (v0.1.28, at Rp 16,000) that the order was built from; `docs/parts.csv` / Notion keep the per-part list.

	Rp	USD	Status
Tokopedia cart: parts (robot, rack, Pi, camera, battery pack, PSU, electronics)	≈ 5,500k	≈ 306	In the cart, not yet paid
Tokopedia cart: shipping to Bali (≈ 30 shops)	≈ 1,000k	≈ 56	Was ≈ Rp 676k to the old Bandung address
Items still to add (`docs/cart-batch-2.md` : corner brackets, inserts, panel lamps, wire, opto board, M3/M2/M4 screws…), incl. shipping	≈ 630k	≈ 35	Priced or estimated
Motion PCB, JLCPCB (5 boards, 2 assembled), FedEx	≈ 576k	32	Ordered and paid (P22)
3D printing, BeauxButton (all printed parts)	2,000k	≈ 111	Quote; shipping free (orders over Rp 1M)
Ballast and epoxy (dummy drives), tags/labels, table mat	≈ 290k	≈ 16	Estimate
Claude Pro, 1 month	360k	20	Estimate
Total	**≈ 10,356k**	**≈ 575**	**≈ 75 over the 500 USD target**

Not counted: possible import duty on the PCB (≈ +10–13 USD). The Delta PLC is not in the budget (loan or sponsor); the ESP32 fallback rack controller is in the cart.

About the 500 USD. It is the team's own target, matching the 500 USD grant each Regional Final team receives. The Delta Cup 2026 guidebook sets no budget cap and no penalty for spending more; it only asks for a component list with estimated costs in the proposal (checked 25 Oct). A clear, itemised budget supports the "technical viability" score.

Why it is over (the plan had ≈ 20 USD of margin, ≈ 480 USD, before v0.1.27; the main additions since are below. They add up to more than the ≈ 95 USD difference because the weaker rupiah (16,000 → 18,000) and actual prices offset part of them):

1. Raspberry Pi 3B+ bought (+70): the team turned out not to own one (23 Oct).
2. ESP32 rack controller (+18): a Delta PLC loan looks unlikely, so the rack needs its own controller.
3. Items the prototype review found missing (+29): spare ESP32 and driver, V-wheel axle bolts, drag chain, crimp tool, wiring extras, tethered power lead.
4. Shipping to Bali (≈ 56): about 30 small shops, each with its own fee.
5. Claude Pro, one month (+20).

Savings already taken: no travel router (–40), panel lamps instead of a tower light (–5), local suppliers instead of AliExpress (no import tax).

Savings plan to get under 500 USD (needs ≈ 75 USD ≈ Rp 1.35M; keeps the Pi and camera):

Tier 1: no downside (≈ Rp 465k, 26 USD)

Cut	Saves
Remove the gantry plate set from the cart (the carriages are printed, so it is unused)	Rp 180k
Remove the spare 4S BMS (the Langit pack has its own)	≈ Rp 50k
microSD from a local shop (≈ Rp 100k instead of Rp 217k)	≈ Rp 115k
Borrow a JST-XH crimp tool from the lab	≈ Rp 120k

Tier 2: small trade-offs (≈ Rp 1.0–1.2M, 55–65 USD)

Cut	Saves	Trade-off
Ship to where the robot is built (e.g. Malang), or merge single-item shops into shops already in the cart	≈ Rp 400–500k	Only if assembly is not in Bali
OV5647 wide-angle camera (Rajacell, Rp 126k) instead of the IMX708 board (Rp 595k)	≈ Rp 470k	Keeps vision; fixed focus, more distortion; confirm FOV ≥ ~120° (the standard 54° board can't see the approach tag); camera slot needs a re-fit
3 panel lamps or the WS2812 LEDs instead of 15 lamps (the shop's minimum is 5 per colour)	≈ Rp 90k	—
Drop the spare ESP32-S3 and the 5th TMC2209	≈ Rp 160k	No spare if one fails
Ask BeauxButton to re-quote with the per-part print settings (light parts 3 walls / 10–15 %)	≈ Rp 200k	Only if the quote assumed 5 walls / 20 % everywhere

Recommended: Tier 1 + shipping + the OV5647 camera → **≈ 498 USD**. If shipping can't change, use the lamps, spares and print re-quote instead → **≈ 495–500 USD**. Check first that the gantry set and the second BMS are in the cart.

Tier 3: if more margin is needed

Cut	Saves	Trade-off
Used Pi 3B, or borrow one from the campus lab	≈ Rp 400k–1.1M	Used-board risk
Cheaper print shop ("its basic", Rp 800/g)	≈ Rp 200–400k	Only worth it if it also ships free
Count Claude Pro (Rp 360k) as a personal tool, not a project part	Rp 360k	Accounting choice, not money saved
No Pi or camera: line following + laptop dashboard	≈ Rp 2.0M	Loses machine vision (a Track 1 focus area); new firmware work before 10 Nov

1.1–1.4 Planning estimate by category (v0.1.28, Rp 16,000 = 1 USD)

Prices are typical AliExpress / local-store (Tokopedia) prices and **must be checked when ordering**. AliExpress parcels over USD 3 pay ≈ 30–40 % Indonesian import tax, so local is preferred unless much cheaper (see `docs/sourcing-options.md`). The live, per-part order list is the Notion **Parts & Order List** database, mirrored in `docs/parts.csv` (IDs P01…); this section keeps the category totals. Shipping is assumed free or small (AliExpress Choice / local).

1.1 Robot — mechanical

Item	Qty	USD
NEMA17 42×40, ≥ 0.4 N·m (Tokopedia)	4	42
2020 V-slot 6 × 1 m (4.96 m of cuts + kerf) + 2040 V-slot 0.42 m (robot + rack + spare station), Lajarus Teknik	—	23
Mini V-wheels + 4 mini eccentric spacers, 4 plain 6 mm spacers, 10 shims 5 × 8 × 1	8	10
Tr8×2 lead screw 350 + anti-backlash nut + coupler + KFL08 flange bearing	1	7
GT2 belt (Y axis), 20T, idler	set	3
Wheels 80 × 24 mm: printed PETG hub + TPU 95A tire (filament in §1.4)	2	0
Ball casters (2 rear + 1 anti-tip), printed spacers	3	3
MG90S servo (Tokopedia)	3	4
Lever microswitches KW11-3Z (12 + 3 spare)	15	3
Fastener kit (~120 M5 T-nuts, screws incl. 7 × M5×16 + 3 × M5×25 for the tray, 8 × M5×30 + nylocks for the V-wheel axles, ~48 corner brackets, M3 + ~60 M3 / ~15 M5 heat-set inserts)	—	30
Cable carrier (drag chain) 1 m, for the Z/Y harness	1	2
Subtotal		**127**

1.2 Robot — electronics

Item	Qty	USD
Raspberry Pi 3 Model B+ (not team-owned after all, Tokopedia)	1	70
IMX708 120° camera board (Camera Module 3 **Wide** class, Tokopedia)	1	37
microSD 32 GB A1	1	8
CSI flex cable 500 mm	1	1
ESP32-S3-DevKitC-1 (Tokopedia), 1 + 1 spare	2	14
TMC2209 StepStick (Tokopedia), 4 + 1 spare	5	15
Buck 5 V/5 A (XY-3606) + 2 × XL4015 CC/CV (charger, 5.8 V servo buck)	3	4
INA226 module + **10 mΩ / R010** 2512 shunt (local modules ship R100: swap it)	1	2
18650 3.5 Ah × 4 + 4S BMS + holder/nickel strip	set	33
XT60/XT30, fuse holder, rocker switch, robot E-stop, silicone wire, JST-XH kit	set	25
Motion PCB, JLCPCB 5 pcs + Economic SMD assembly of 2 + shipping	1	40
Passives, headers, StepStick sockets, TVS, caps, WS2812, buzzer (itemised, Tokopedia)	set	5

Item	Qty	USD
USB-A to USB-C cable, 90° C plug (Pi → DevKit)	1	2
VMOT relay for the E-stop (12 V automotive, 30/40 A) + flyback diode	1	2
Harness extras: Pi micro-USB pigtail, servo/stepper extension leads, heat-shrink, Pi heatsink	set	6
JST-XH crimp tool (skip if the team has one)	1	7
Subtotal		**271**

1.3 Rack & spare station — electrical

Item	Qty	USD
Delta 24 V DIN-rail PSU, DRL-24V120W1EN (120 W)	1	39
DIN rail, terminal blocks, 24 V relay module, UK5-HESI fuse terminal + 3.15 A fuse (CNC Store Bandung)	set	5
Status lamps: 3 × 22 mm 24 V panel indicators (R/Y/G) in a 3-hole box (replaces the tower light, v0.1.27)	1	3
Rack E-stop (2 NC contacts)	1	3
Dock contacts (pogo/brass), copper sheet, spare-station cables (4-core signal + 2-core dock power)	set	9
Delta DVP-12SE11T PLC	1	**not in budget, loan/sponsor** (≈ 150–250)
ESP32 rack controller (PLC fallback, contingency; likely needed): DevKit + 8-ch 24 V opto inputs + 2 × dual MOSFET outputs + buck (02 §4.4)	set	18
Tethered power lead, 2 m + XT60 (battery fallback, R7)	1	2
Subtotal		**79**

1.4 Printing, demo materials & services

Item	USD
PETG 2 kg	36
PLA 1 kg	15
TPU 95A 0.25 kg	8
Steel ballast + epoxy (dummy drives)	8
Tag/label printing, table mat	10
Claude Pro, 1 month (Rp360k)	23
Subtotal	**100**

1.5 Total (planning estimate; the forecast is in §1.0)

	USD
Robot mechanical	127
Robot electronics	271
Rack electrical	79 (61 with the PLC loan)
Printing, demo & services	100
Total	**≈ 577** at Rp 16,000 (of 500: ≈ 77 over). Superseded by the §1.0 forecast: **≈ 575 at Rp 18,000 from the actual cart and quotes**

v0.1.28: ≈ 548 → ≈ 577 USD (≈ 77 over). The prototype review (`docs/review/Corvin-project-review.pdf` , 23 Oct) found missing or short items: a spare ESP32-S3 DevKit (+7) and a 5th TMC2209 (+3, the shopping list already bought 5), M5×30 V-wheel axles + nylocks and the belt-clamp M3 (+2), a drag chain (+2), harness extras (+6), a JST-XH crimper (+7) and a tethered power lead (+2). Not counted yet: import duty on the motion PCB (≈ +10–13), and the print-service quote vs the filament lines below. **v0.1.27: the budget was ≈ 48 USD over** after buying the Raspberry Pi (+70), one month of Claude Pro (+23) and counting the ESP32 rack controller (+18), since PLC access now looks unlikely; dropping the travel router (–40, the Pi is the access point) and replacing the tower light with panel lamps (–5) took back 45. The savings options below recover ≈ 60–90 USD; otherwise the overrun needs the team's approval. Before that, the margin was ≈ 20 USD after the v0.1.12–v0.1.17 move to local (Tokopedia) suppliers, buying the travel router (v0.1.16) and JLCPCB SMD assembly of the motion board (v0.1.20); only the motion PCB (JLCPCB) is imported. **Savings options** (use them if prices come in higher, or a router must be bought): wide-angle OV5647 camera instead of the IMX708 board (–29, more distortion) · team-owned filament or fasteners (–20 to –50) · drop the 3rd servo and spare switches (–5). Priced backup options and local (Tokopedia) sources: `docs/sourcing-options.md` . **PLC**: request a loan from Delta Thailand this week. If none, buying a DVP12SE11T costs ≈ Rp2.6–3.3M (≈ 165–210 USD) on Tokopedia, which does not fit the budget, so use the ESP32 fallback rack controller (≈ +20 USD, see 02 §4.4) unless a sponsor covers it. **Pi 3**: not team-owned after all (23 Oct), so a 3B+ is bought (≈ Rp1.1M ≈ 70 USD, v0.1.27).